

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Provisioned and operated multi-cloud GPU clusters (AWS, GCP, CoreWeave) with NVIDIA/CUDA hardware and InfiniBand/RoCE networking, supporting training and high-availability inference on shared fleet capacity.
- Built a deployment-gated evaluation framework enforcing guardrails and quality/regression checks across model outputs, blocking 15% of failed releases before traffic cutover.
- Engineered a pluggable distributed executor routing workloads across Docker and AWS Lambda with an S3 artifact store, eliminating redundant recomputation under scarce-capacity constraints.
- Own production services end-to-end, architecture through deployment, monitoring, and on-call, handling millions of daily transactions at 99%+ uptime.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Extended and optimized inference framework internals (vLLM-class serving) with kernel-level development (CUDA/Triton), distributed KV cache, and speculative decoding to improve LLM serving performance.
- Cut p99 inference latency 420ms to 140ms across 50K+ daily requests by migrating synchronous serving to async vLLM with autoscaling.
- Ran model training and fine-tuning on multi-GPU clusters with compression and quantization for efficient inference, and built evaluators tracking drift and regression across model versions.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Built and operated 25+ production microservices in Java, C++, and SQL-backed distributed systems for 1M+ users, sustaining 99.9% uptime over 3 years on AWS (EC2, RDS, SQS).
- Standardized failure-handling contracts across 25+ financial services, reducing production escalations 60% in one quarter via consistent retry, timeout, and propagation semantics.

PROJECTS

Distributed KV Store / VPC Control Plane Simulator

github.com/ganeshhgupta/distributed-kv-store

- Raft-consensus distributed KV store handling concurrent writes with linearizable reads and log compaction; companion VPC control plane simulator modeling subnet routing and fleet-level resource placement.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- Production LLM eval and observability platform with hallucination detection, drift tracking, rubric-based scoring, and regression alerting across model versions; FastAPI backend, PostgreSQL.

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- Multi-agent system automating production incident diagnosis via tool-use chains; routes alerts to specialized sub-agents and synthesizes evidence-backed root cause reports.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: Python, Go, Rust, C++, Java, SQL

Model Serving & Inference: vLLM, CUDA/Triton Kernel Development, Distributed KV Cache, Speculative Decoding, Quantization, Model Compression

ML Infra: Model Training/Fine-Tuning (Multi-GPU), Evaluation & Regression Frameworks, Deployment Gating, Observability

Infrastructure: Multi-Cloud GPU Clusters (AWS, GCP, CoreWeave), InfiniBand/RoCE, Distributed Systems, Docker, Kubernetes, CI/CD

CERTIFICATIONS

AWS Certified Developer Associate | Google Cloud Professional Machine Learning Engineer